

Federated TimeVQVAE Anomaly Detection: Prior Sharing Requires an Aligned Tokenizer

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Abstract—Time series anomaly detection is often carried out by parties that cannot pool their raw data. We ask what should be shared when the detector is a two-stage generative model: TimeVQVAE-AD first learns a discrete tokenizer of the signal, then a prior over the resulting tokens. The two stages cannot be federated independently. Sharing the prior helps only once the clients’ tokenizers have been aligned by averaging their encoders, and the identical change is a null when they have not been, because the prior is indexed by token identity and unaligned clients use those identities differently. On the UCR anomaly archive, with every series split across five clients, the federated model improves over local training in the ranking of anomalous timesteps and matches pooled training on the archive’s own localization metric on most series, a metric on which pooled training, which sees all the data, does not separate from local training either. We also describe a case in which averaging spreads a false alarm raised by a minority of the clients to all of them, and show that fusing the local models’ scores instead of their weights avoids it.

Index Terms—federated learning, time series anomaly detection, vector quantization, generative models, model aggregation

I. INTRODUCTION

Time series anomaly detection [1], [2] is often performed where sensors, machines or organizations collect data independently and cannot share their raw observations [3], [4], and federated learning (FL) [5] exploits that data without centralizing it. On the UCR anomaly archive [6] the collaboration is worth having: whenever the methods can be distinguished, pooled training outperforms the average client, although not necessarily the best one. We study how to realize it for TimeVQVAE-AD [7], a two-stage generative detector composed of a VQ-VAE [8] tokenizer and a masked prior over the resulting discrete tokens [9], whose structure raises a question that is less explicit in end-to-end detectors: which components should be shared, and how should federation of the tokenizer and of the prior be coordinated?

These decisions turn out not to be separable. Sharing the codebook alone can underperform purely local training, while federating the encoder, pooling batch normalization (BN) statistics and aggregating the codebook through sufficient statistics does not improve on the local baseline when the prior stays local. The strongest configuration combines all of these with a mostly shared prior, keeping the channel embedding, output bias and decoder local; it matches centralized training exactly on three of the four development series on

which the methods separate at all, and fails on the fourth, `ucr_170`, which we analyze as a failure case. On a pre-registered random sample of 50 further series it improves the ranking of anomalous timesteps over local training on 32 of 48 ($p = 0.029$ on the area under the precision-recall curve, AUPRC) and matches centralized training on the archive’s own accuracy on 37 of 48, a metric on which the pre-registered primary endpoint reaches significance for no contrast, centralized training included, so that what the sample establishes is parity with the pooled upper bound rather than a gain in localization.

Matched ablations attribute the improvement to prior sharing, but only after the token representation has been aligned across clients, consistently with the observation that parameter averaging requires compatible internal representations [10], [11]. For a discrete generative model the requirement is sharper: prior parameters can be shared only when token indices have common semantics *and* clients map windows to those indices consistently. We contribute a coordinated federation strategy for such detectors, in which alignment is a measured property of the checkpoints rather than a label, and a case study of a failure mode of weight-space aggregation together with a function-space repair that recovers it where it occurs.

II. RELATED WORK

Federated learning. FL lets clients train a shared model while their data stay local [15], with FedAvg as the reference point [5]. Parameter averaging is brittle when local models drift apart, and the remedies act on the local objective, the local update or the server [16]–[18], or match internal units before averaging them [11]; a second line shares only part of the model, a global body with personal heads [19] or the mirror arrangement [20]. In federated time series anomaly detection, Fed-ExDNN aggregates per-device exemplar modules through constrained clustering, aligning them before averaging [3], and PeFAD adapts a pre-trained language model, distilling through a shared synthetic dataset [4]; in both the aggregated representation is the terminal one, read directly by a scoring function. To our knowledge no federated work addresses a pipeline in which a discrete tokenizer produces the symbols consumed by a second, generative prior.

Discrete tokenization and shared codebooks. VQ-VAE builds a discrete latent representation on a learned codebook

and fits a prior over the codes in a second stage [8]; TimeVQ-VAE carries this to time series with a Transformer prior [9] and TimeVQVAE-AD reuses the structure for detection [7]. The closest analogues to sharing a codebook transfer knowledge without averaging full weight vectors: federated clustering of centroids [21], shared prototypes [22], distributed clustering for globally consistent representations [23]. None of them is a discrete vocabulary read by a downstream model, so aligning them improves the quality of the aggregate, not its well-definedness. When the vocabulary is itself federated, alignment becomes a precondition for prior sharing to be well posed at all.

III. BACKGROUND: TIMEVQVAE-AD

We summarize only what the federation design needs; for the method itself we refer to the original paper [7]. Throughout, a *client* owns a set of windows of a univariate series and never shares them.

Stage 1 is a VQ-VAE [8] over the short-time Fourier transform of a window: an encoder E and a vector quantizer with codebook $\mathcal{C} = \{e_k\}_{k=1}^K$ map $x \in \mathbb{R}^T$ to a grid $s \in \{1, \dots, K\}^{H \times W}$ of code indices, which we call *tokens* and a decoder reconstructs; the convolutional kernels are frequency-independent, so each token localizes a specific time and frequency region. Stage 2 freezes E and the decoder and fits a MaskGIT prior [24] $p_\theta(s)$ by masked modeling, as in TimeVQVAE [9]. The second consumes what the first produces: the prior is a categorical model *over the indices* the first stage emits, and the rows of its output layer are indexed by that stage's codebook.

Detection uses no label. A masking window of odd latent width ℓ slides along the temporal axis of the latent grid, masking whole columns across all frequency bands, and the score of a latent step is the mean negative log-likelihood the prior assigns to the true tokens of the masked region. The maps obtained at $\ell = \text{odd}(rW)$ for $r \in \{0.1, 0.3, 0.5\}$ are summed, averaged over frequency, mapped back to data space, accumulated over the rolling windows and averaged with their centered moving average, giving one score per timestep. Two consequences matter for federation: the score comes from the prior alone, so any choice that touches the tokenizer reaches detection only through the token grid the prior consumes and the vocabulary it predicts, and the decoder takes no part in it.

Finally, the coupling between the stages restricts what federation can mean. A token is a pointer into a codebook, so averaging prior parameters presupposes that index k denotes the same codeword on every client; and since the index is produced by E and the quantizer jointly, it further presupposes that clients agree on *which* windows map to k . Sharing a dictionary secures the first condition and not the second.

IV. FEDERATION DESIGN

Federating TimeVQVAE-AD requires coordinating the tokenizer, which fixes the indices assigned to each window, and the prior, which models their distribution, so we treat the encoder, the BN statistics, the codebook and the prior

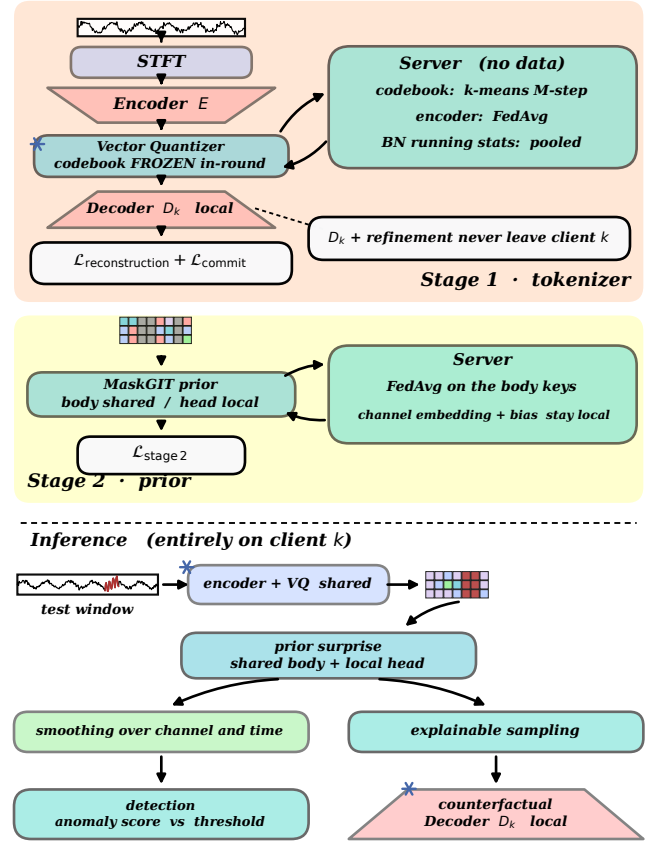


Fig. 1. Configuration (g) on client k . Stage 1 federates the encoder and its BN running statistics and keeps the broadcast codebook fixed within a round, the server updating it from (2); the decoder is never sent. Stage 2 shares the body of the MaskGIT prior, keeping the channel embedding and output bias local. Inference is local and forks into the anomaly score and the counterfactual.

separately; the decoder always stays local, as it does not contribute to the score. Client j holds n_j training windows and contributes with weight $\omega_j = n_j / \sum_i n_i$; raw windows never leave the client. Fig. 1 shows the arrangement for configuration (g) below.

A. Tokenizer federation

The encoder can remain local or be shared through FedAvg [5]; when it is federated, its BN running statistics are pooled:

$$\bar{\phi} = \sum_j \omega_j \phi_j, \quad \bar{\mu} = \sum_j \omega_j \mu_j, \quad \bar{v} = \sum_j \omega_j (v_j + \mu_j^2) - \bar{\mu}^2. \quad (1)$$

This is the opposite of the FedBN prescription [25]: here the clients observe the same sensor and regime, so there is no feature shift for a local BN layer to absorb, while a shared tokenizer requires that every client normalize identically before quantization.

The codebook can be federated in two ways. Direct FedAvg averages corresponding codewords after local training, which need not leave the average near any of them once the clients'

quantizers have drifted apart during the round. Our alternative keeps the broadcast codebook fixed during the round and aggregates assignment statistics: for every code k , client j sends its assignment count $c_{j,k}$ and the sum $m_{j,k}$ of the corresponding pre-quantization encoder vectors, and the server computes

$$N = \sum_j c_j, \quad M = \sum_j m_j, \quad e_k = M_k / \tilde{N}_k, \quad (2)$$

with $\tilde{N}_k$ a Laplace-smoothed count, so that each global codeword is updated from statistics accumulated across clients without transmitting latent samples or raw data. This motivates the alternative but is not a demonstrated defect of averaging: Section VI-D finds the two rules indistinguishable in aggregate. Fixing the codebook within a round does not freeze the tokenizer, since the encoder keeps changing.

B. Prior federation

Prior federation requires a shared vocabulary: the MaskGIT prior predicts codebook indices, so index k must refer to the same codeword across clients, and averaging prior parameters learned with independent dictionaries would combine parameters attached to unrelated symbols. We therefore allow prior sharing only when clients use a common codebook. A common dictionary alone does not guarantee identical assignments, which also depend on the encoder, so we evaluate prior sharing both with local encoders and after encoder federation.

We compare a fully local prior with a partially shared one, in which the transformer, the token embedding, the frequency and time embeddings and the readout are shared while the channel component of the positional encoding and an additive output bias remain local; on univariate data personalization therefore acts mainly through the output bias. Stage 2 is aggregated after a fixed number of local epochs, or, in a variant, after exactly τ optimizer steps.

C. Evaluated configurations

Table I summarizes the configurations. Two matched comparisons carry the argument: (c) and (e) differ only in prior sharing while the encoder stays local, and (f) and (g) make the same comparison after encoder federation. The last two rows replace one stage by its pooled counterpart, so that the two stages can be blamed separately. Configurations (j) to (m) vary how the encoder is aggregated at the codebook and prior settings of (f), with FedProx [16] acting on the local objective, FedProto [22] broadcasting per-code prototypes, and (m) sharing only the starting point.

D. Score fusion

As an alternative to parameter sharing we also combine independently trained local models at inference time, as outlier ensembles [12], [13] and function-space federated aggregation [14] do: each client keeps its tokenizer and prior unchanged, computes its profile and standardizes it over time, and the final score is the pointwise mean across clients, so that only per-timestep score vectors are combined and no alignment

TABLE I
CONFIGURATIONS EVALUATED. “FEDAVG” IS COUNT-WEIGHTED PARAMETER AVERAGING, “POOLED” THE AGGREGATION OF BN STATISTICS, “SUFF. STAT.” THE CODEBOOK UPDATE OF (2), “PARTIAL” THE SHARED PRIOR WITH CLIENT-SPECIFIC COMPONENTS LOCAL. n = DEVELOPMENT SERIES RUN AND, IN PARENTHESES, COVERAGE OF THE FOUR THAT SEPARATE THE ARMS AT ALL (SECTION VI-A), SO 4/4 IS A COMPLETE COVER.

Configuration	Encoder	BN	Codebook	Prior	Stage 2	n (disc.)
(a) local	local	—	local	local	—	10 (4/4)
(b) centralized	one model on the pooled windows (reference)					10 (4/4)
(c) codebook only	local	—	suff. stat.	local	—	10 (4/4)
(d) codebook only, averaged dictionary	local	—	FedAvg	local	—	10 (4/4)
(e) codebook and shared prior body	local	—	suff. stat.	partial	epochs	10 (4/4)
(f) aligned tokenizer, local prior (A1)	FedAvg	pooled	suff. stat.	local	—	10 (4/4)
(g) aligned tokenizer, shared prior body (A2)	FedAvg	pooled	suff. stat.	partial	epochs	10 (4/4)
(h) (g) with a step-counted rhythm	FedAvg	pooled	suff. stat.	partial	τ steps	10 (4/4)
(i) (h) with a mobile dictionary	FedAvg	pooled	FedAvg	partial	τ steps	3 (3/4)
<i>Encoder aggregation rule, isolated at a local prior^b</i>						
(j) encoder local BN stats	FedAvg, FedAvg	affine	suff. stat.	local	—	10 (4/4)
(k) (j) with FedProx	FedProx	affine	suff. stat.	local	—	10 (4/4)
(l) (j) with FedProto	prototypes	init	suff. stat.	local	—	10 (4/4)
(m) (j) with a shared initialization only	init	init	suff. stat.	local	—	10 (4/4)
(n) score fusion ^a	local	—	local	local	—	10 (4/4)
<i>pooled tokenizer^c</i>	<i>pooled stage 1, federated stage 2</i>					1 (1/4)
<i>pooled prior^c</i>	<i>federated stage 1 of (g), stage 2 on pooled tokens</i>					4 (4/4)

^aTrained exactly as (a); only inference differs. ^bRows (j) to (m) target the same 87 encoder tensors as (f) but keep the BN running statistics local (“affine” shares BN scale and shift every round, “init” only at initialization), so they are matched to one another and *not* to (f), (g) and (h): their baseline is (j). FedProx uses $\mu = 0.01$; (m) is (l) with the prototype pull set to zero. ^cNot federation-legal, diagnostic only. *Pooled prior* runs on exactly the four discriminating series, so its $n = 4$ covers the informative subset completely. Both are matched to the reference of Section VI-C, not to (g).

of parameters, codewords or indices is needed. No training is shared, and the profiles can be averaged only because the split divides the training segment while the test segment is common, so they are defined on the same timesteps; all five models must also be present at inference. Neither holds where clients score their own streams. We also report the median, the trimmed mean and the mean of normalized ranks.

V. EXPERIMENTAL PROTOCOL

A. Benchmark, choice of series and split

Two settings are inherited from the original method rather than tuned, since it states them as such: the window length is $T = 2P$, twice the measured period of the series, and each input window is z-normalized before the transform; we also keep its frequency resolution ($H = 3$ bands) and latent temporal width (32). With $T = 2P$ the UCR Time Series Anomaly archive [6] leaves 180 usable series: the remainder either have no measured period or leave the smallest client with fewer samples than one window. Ten form a development set on which the configurations were explored, selected by period and domain rather than by detection performance, the period-

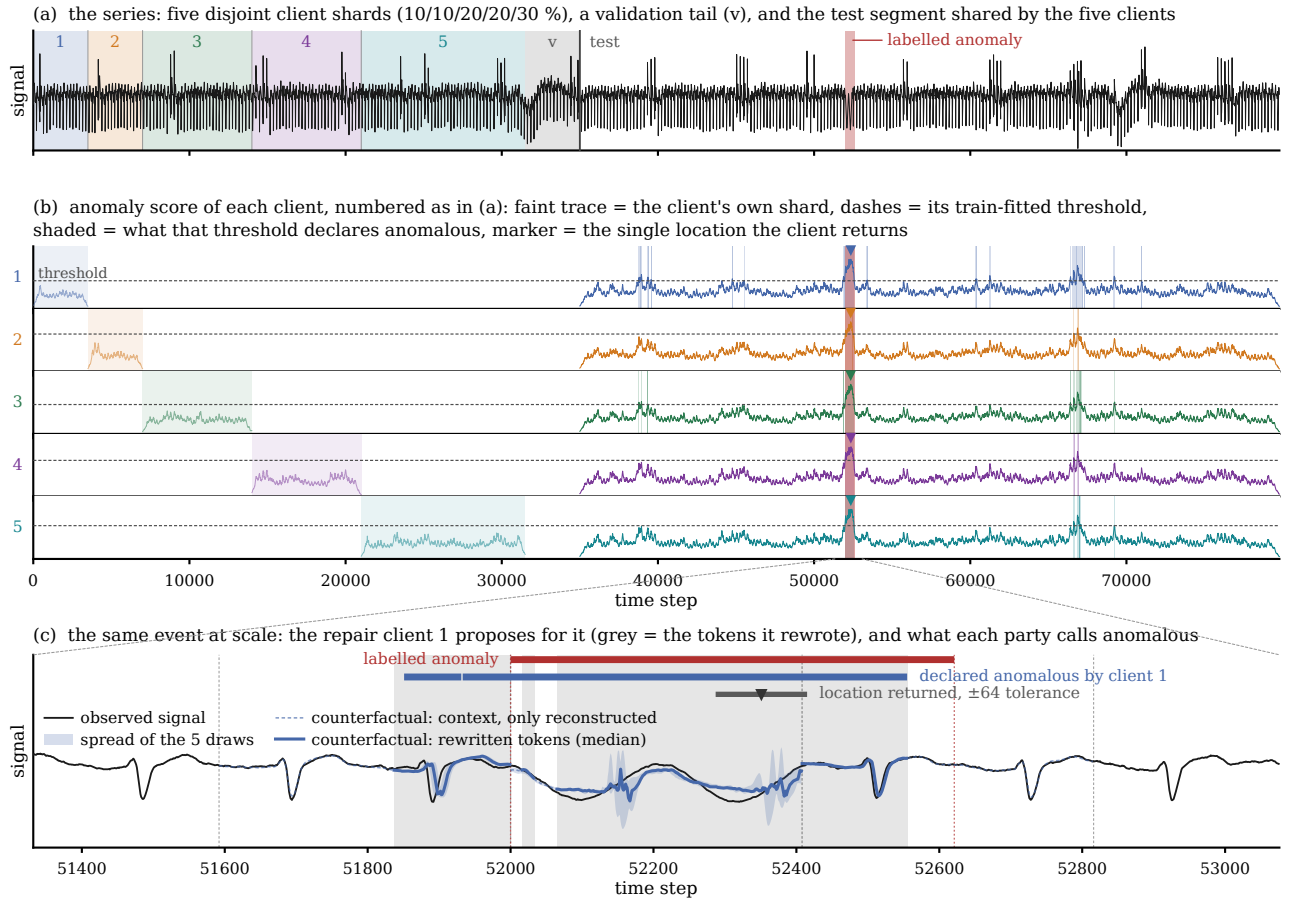


Fig. 2. One development series, `ucr_001`, under configuration (g). (a) the training segment is cut chronologically into five client shards of 10/10/20/20/30% plus a validation tail (v); the test segment and its labels are shared and never seen in training. (b) each client’s per-timestep score, its own train-fitted threshold (dashed), the timesteps it declares anomalous (tinted) and the single location it returns under the archive protocol (marker): all five agree, inside the labelled event, while their thresholds admit 1.2 to 4.1% of the test timesteps. (c) the same event with the explanation, built with *no label in the loop*: the timesteps above client 1’s threshold are masked across frequency, resampled from the prior with the remaining tokens as context and decoded locally (band = five draws, solid = median). Every configuration already localizes this series.

aligned window length being approximately logarithmically spaced from 182 to 970 samples. A disjoint set of 50 was then drawn uniformly at random with a fixed seed from the remaining 170, the same draw fixing their execution order; the sample, its order, the three configurations run on it, the primary metric, the stopping rule, a minimum sample size and a freeze date were all fixed before observing any of its results (Section VI-F).

Each series is an independent federated task: no parameter, statistic or window is shared between series. Within a series the training segment is divided chronologically among five clients holding 10%, 10%, 20%, 20% and 30% of the data, the final 10% is held out for validation, and the archive’s test segment and labels are never used in training (Fig. 2). Windows are extracted with unit stride, so a shard of length L contributes $L - T + 1$ windows, and these counts are the aggregation weights of Section IV. The split is deliberately homogeneous: the five clients observe the same sensor, regime and units and differ only in how much of the series they hold.

This removes client heterogeneity as the intended explanation for differences between local and federated training.

B. Training

All configurations use the same trainer and select the best validation round, but the stopping criterion is granular to the setting: local and centralized training stop on validation patience counted in optimizer steps inside a fixed step budget, federated configurations on patience counted in aggregation rounds, validation being defined once per round; a federated run that exhausts its round ceiling without triggering patience is marked truncated and excluded. The baselines’ step budget binds in 9% of their stage-runs on the development set and in 15 to 25% on the confirmation sample, and those runs are retained, so the baselines are, if anything, the arm more likely to be reported before convergence.

Capacity is fixed across configurations: a codebook of $K = 64$ entries of dimension 64, an encoder of base width 16, and a four-layer prior with four heads and width 128, optimized

with AdamW. The codebook is smaller than in the original method, so our reference results are not directly comparable with its published accuracy. One seed is run per experimental cell. We also report a parameter-free floor, the squared residual of a centered moving average of length 10 pushed through the same detection path; having no parameters it admits no federated variant and is a lower reference, not a competitor.

C. Metrics and unit of analysis

We report the archive’s accuracy, where a method returns one predicted location per series and is correct within 64 timesteps of the labelled anomaly, and the AUPRC of the per-timestep score. The two are not interchangeable: the archive metric is the benchmark’s own, so it leaves us no discretion, and it is correspondingly coarse, one hit or miss per series; AUPRC is ours and uses the whole profile. We do not report the area under the ROC curve (AUROC), which saturates here (it exceeds 0.95 on 8 of 48 confirmation series for a configuration whose predicted location misses the anomaly entirely), nor threshold-based scores: the train-fitted threshold, a sum over rates of per-rate train quantiles, leaves nothing of the training profile above it at the nominal $q = 0.99$ and admits between 0.000 and 0.978 of the test timesteps depending on the arm, nor range-aware measures [26], whose scale depends on anomaly duration. The unit of analysis is the series: the five clients are shards of one signal, not independent replicates, so comparisons are paired by series.

VI. RESULTS

Results are from the ten-series development set unless stated otherwise, one seed per cell, with client-level results averaged to one value per series.

A. Which series can separate the methods

Local training already scores 1.00 on `ucr_001`, `ucr_083` and `ucr_086`, leaving no room for improvement, and both local and centralized training score 0.00 on `ucr_082`, `ucr_222` and `ucr_229`, which therefore carry no evidence about federation. The remaining four, `ucr_011`, `ucr_014`, `ucr_043` and `ucr_170`, are the informative cases for the archive metric, and on four series a two-sided sign test cannot reach $p < 0.05$ (its minimum attainable value is 0.125), so we emphasize per-series magnitudes and win/loss counts rather than significance.

B. Reference points

Pooling all client windows improves over the average local model on 9 of 10 series in AUPRC and on all four discriminating series in the archive metric, where local training obtains 0.40, 0.80, 0.20 and 0.20 against 1.00 for centralized training in every case. Against the best individual client the picture differs: centralized training exceeds it on 7 of 10 series in AUPRC ($p = 0.34$), but on the archive metric never detects an anomaly that no individual client detects, the two scoring identically on all ten series, and $\max_j$ is an oracle either way. The parameter-free floor stays far below all of them, median

AUPRC 0.007 against 0.353 for local and 0.899 for centralized training, and is exceeded by every trained arm on all ten series, though on `ucr_083` it reaches 0.825, consistent with that series being uninformative about federation.

C. Main federated configuration

Configuration (g) reaches the centralized archive score on three of the four discriminating series and fails on the fourth (Table II): three improvements, one degradation and six ties, all six on the non-discriminating series. In AUPRC it improves nine series and degrades one, the gains concentrated on three (+0.42, +0.46, +0.30) and six of them below 0.036. Against centralized training it reaches the same archive score on 9 of 10 series and falls below only on `ucr_170`, while in AUPRC it stays below on 8 of 10: equal localization does not imply equal ranking of the full profile.

The comparison with local training depends on where local training is stopped. Against the baseline used throughout, which stops on validation patience, federation improves 9 of 10 series in AUPRC (sign test $p = 0.021$); against a baseline trained to a budget four to five times larger with early stopping disabled, the median difference is essentially unchanged (+0.022 against +0.021) but the count falls to 6 against 4 ($p = 0.75$), the longer-trained baseline being not uniformly better but more variable: federation improves local training stopped at its own criterion, not at any budget.

The stage-2 synchronization rhythm, by contrast, is a null on both sets. Replacing epoch-based rounds with $\tau = 64$ optimizer steps at an identical stage 1 checkpoint improves four development series and degrades six (median -0.001); the reference there is not the (g) of Table II but the arm sharing (h)’s low-noise stage-2 validation oracle, frozen masks in fp32, which small τ requires and which changes the round selected, not the trajectory. On the confirmation sample that arm was not run, so (h) is compared with (g) and the oracle travels with τ : from the bit-identical stage 1 of (g) it wins 25 against 23 in AUPRC (median 0.0000, $p = 0.89$) and 2 against 4 with 42 ties on the archive metric ($p = 0.69$). Markedly less frequent aggregation, the same reference at 31 local epochs per round with ≈ 310 local steps between server updates instead of 100 to 190, costs 0.198 and 0.245 AUPRC on `ucr_011` and `ucr_043`, so the default frequency is sufficient and increasing it buys nothing.

D. Where the improvement comes from

To isolate prior sharing we apply the same modification under two tokenizer configurations: (c) versus (e) with a shared codebook but local encoders, and (f) versus (g) after encoder federation, the set of shared prior tensors being identical in the two (55 keys, with the channel embedding and output bias local). With the aligned tokenizer, prior sharing improves 9 of 10 series in AUPRC (median +0.069) and all four discriminating series in the archive metric; with local encoders it gives 4 improvements and 6 degradations (median -0.001) and splits 2 against 2. The difference is positive on 8 of 10 series in AUPRC (median +0.136) and on all five non-tied

TABLE II

DEVELOPMENT SET. LEFT: ARCHIVE ACCURACY AT TOLERANCE 64; RIGHT: AUPRC OF THE PER-TIMESTEP SCORE, WITH THE PARAMETER-FREE FLOOR AS LEFTMOST REFERENCE. $\max_j$ IS THE BEST INDIVIDUAL CLIENT AND REQUIRES TEST LABELS TO SELECT. THE FOUR DISCRIMINATING SERIES ARE ABOVE THE RULE. ON `ucr_082` EVERY ARM IS AT FLOOR: THE COUNTS IN THE TEXT REST ON DIFFERENCES IN THE FIFTH DECIMAL.

series	accuracy@64				AUPRC			
	local	$\max_j$	centr.	(g)	floor	local	$\max_j$	centr. (g)
<code>ucr_011</code>	0.40	1.00	1.00	1.00	0.092	0.403	0.997	0.976 0.820
<code>ucr_014</code>	0.80	1.00	1.00	1.00	0.003	0.639	0.902	0.964 0.939
<code>ucr_043</code>	0.20	1.00	1.00	1.00	0.046	0.181	0.819	0.875 0.637
<code>ucr_170</code>	0.20	1.00	1.00	0.00	0.015	0.303	0.588	0.878 0.086
<code>ucr_001</code>	1.00	1.00	1.00	1.00	0.007	0.820	0.903	0.920 0.851
<code>ucr_083</code>	1.00	1.00	1.00	1.00	0.825	0.922	0.983	0.932 0.932
<code>ucr_086</code>	1.00	1.00	1.00	1.00	0.007	0.938	0.979	0.986 0.973
<code>ucr_082</code>	0.00	0.00	0.00	0.00	0.000	0.000	0.000	0.000 0.000
<code>ucr_222</code>	0.00	0.00	0.00	0.00	0.002	0.046	0.077	0.083 0.054
<code>ucr_229</code>	0.00	0.00	0.00	0.00	0.002	0.002	0.003	0.018 0.005

series in the archive metric: prior sharing is beneficial only after encoder alignment.

“Alignment” is not a label we attach to the federated arm: it is a property of the checkpoints. We encode the same validation windows with each client and measure how often two clients assign the same latent position to the same code, corrected for chance agreement. Under (g) the five clients tokenize *identically*, $\kappa = 1.00$ on every one of the ten series, whereas under (e), whose encoders are local, the median is $\kappa = 0.07$. The codebook is bit-identical across clients in both arms, so what separates them is not the vocabulary but whether they use it the same way, exactly the condition a prior indexed by code identity requires. The same measurement locates the intermediate mechanisms: encoder FedAvg alone reaches $\kappa = 0.77$ under the BN regime of (j), and pooling the BN running statistics, the only remaining difference, carries it to 1.00, while (m), which shares the initialization and never averages again, sits at $\kappa = 0.04$, indistinguishable from (e) at 0.07. A common starting point does not survive local training; federating the encoder does not make the tokenizers better, it makes them agree.

What is shared matters more than how it is averaged. Relative to (j), the same FedAvg under the same BN regime, replacing encoder FedAvg by FedProx, by FedProto or by a control sharing only the initialization moves the median AUPRC by -0.004 (3 wins), -0.004 (4 wins) and -0.000 (4 wins): even the control that never communicates ties both algorithms. The codebook rule is equally uninformative in aggregate but for the opposite reason: (c) against (d) is five wins each with median $+0.000$ AUPRC, yet the per-series differences (c) $-$ (d) run from -0.946 to $+0.292$, and after alignment the mobile dictionary of (i) loses 0.063 and 0.375 AUPRC against (h) on `ucr_011` and `ucr_043` while gaining 0.270 on `ucr_170`, the one series on which the frozen dictionary is the binding constraint. It is a null of large differences that cancel, so we fix it to sufficient statistics and do not present that choice as a contribution.

E. Failure case and score fusion

`ucr_170` is the only discriminating series on which the main federated configuration fails to recover the local baseline. The five local models obtain AUPRC 0.004, 0.223, 0.329, 0.588 and 0.370, with mean pairwise score-profile correlation 0.56; after federation they become 0.087, 0.094, 0.088, 0.073 and 0.088 while the correlation rises to 0.98. The clients become far more similar, but the common profile ranks a false alarm above the true anomaly, which the centralized model contains as well and ranks below it. The diagnostic configurations locate the responsibility: replacing only the prior of (g) by one fitted on the pooled token grids raises AUPRC to 0.948, 0.947, 0.900 and 0.202 on the four discriminating series against 0.870, 0.944, 0.746 and 0.082 for the matched federated reference, reaching the centralized level everywhere but on `ucr_170`; both diagnostics share the oracle of Section VI-C, so their matched reference is that arm and not the (g) of Table II. Replacing instead only the tokenizer by a pooled one recovers `ucr_170` to 0.578: on most series the binding constraint is the prior, on `ucr_170` the order reverses. Both rest on at most four series and one seed and are bounds, not methods.

Combining the original local models at the score level does much better here. Averaging their standardized profiles gives AUPRC 0.698 against 0.086 for the weight-aggregated model trained on the same shards, and above the best individual client at 0.588. Across the ten series the fusion beats the per-client mean on 7 (median $+0.027$) but the best client only here, where the clients are the least correlated of the ten (0.56 against a median 0.86). Against (g) it loses 3 to 7 in AUPRC (median -0.006), the large losses on the three where prior sharing pays (-0.53 , -0.39 , -0.23), and is level on the archive metric (one win, one loss, eight ties), the endpoint that separates no pair of arms. The gain depends on the rule and lives in the amplitude of the peaks rather than their ordering: the trimmed mean gives 0.690, while the median and the mean of normalized ranks fall to 0.363 and 0.277. Applying it to the federated clients changes nothing (0.087 against 0.086), their profiles being already nearly identical. Fusion is therefore a repair for this failure mode, not a detector that supersedes federation.

F. Confirmation on the random sample

The pre-registered sample tests whether the surviving claims hold out of sample, and its analysis was fixed before any of its numbers existed: the primary endpoint is the archive accuracy at tolerance 64, compared by a paired sign test with ties counted as ties, and AUPRC is secondary. Three configurations are run on it: local, centralized and (g).

Of the fifty series, 48 are analyzed: on `ucr_190` and `ucr_240` the federated arm aborted on non-finite entries in the aggregated codebook, a non-finite latent entering the sufficient statistics of (2) and one coordinate propagating through the one-hot reduction to almost every codeword. It is not codebook collapse, and the failure tracks half-precision

TABLE III

CONFIRMATION SAMPLE, $n = 48$, PAIRED BY SERIES. PRIMARY ENDPOINT: ARCHIVE ACCURACY AT TOLERANCE 64; AUPRC SECONDARY. p IS A TWO-SIDED SIGN TEST WITH TIES EXCLUDED. THE MEAN IS GIVEN ALONGSIDE THE MEDIAN BECAUSE ON THE PRIMARY ENDPOINT THE MEDIAN IS ZERO WHEREVER TIES DOMINATE.

contrast	accuracy@64 (primary)				AUPRC (secondary)			
	W/L/T	med.	mean	p	med.	mean	p	
(g) – local	11/6/31	0.000	+0.083	0.332	+0.027	+0.073	0.029	
centralized – local	14/5/29	0.000	+0.142	0.064	+0.053	+0.141	0.002	
(g) – centralized	3/8/37	0.000	−0.058	0.227	−0.009	−0.067	0.060	

training exposure rather than detection difficulty. The pre-registered rule discards a series from *all* arms when any arm is incomplete; both sit at floor for the reference arms on the primary endpoint, and recovering both as wins would move the primary contrast against local training only from $p = 0.33$ to $p = 0.17$.

Reported in the fixed order, primary endpoint first: *none of the three contrasts reaches $p < 0.05$ on the pre-registered primary endpoint* (Table III). The reason is in the tie counts rather than in the effect: a series contributes only when the arms disagree about whether the single predicted location is found, and they agree on 31 of 48 series for the first contrast and 37 of 48 for the third, so the effective sample size is 17 and 11, not 48. This is a property of the endpoint and not of federation, and the pooled model is the evidence: centralized training sees every client’s data and still does not separate from local training here.

On the secondary endpoint the picture is the one the development set showed, and it is significant: in AUPRC centralized training improves on local training on 35 of 48 series ($p = 0.002$) and (g) on 32 of 48 ($p = 0.029$), while (g) remains below centralized training on 31 of 48 ($p = 0.06$). The tension between the two metrics is therefore not an artifact of a purposive choice of series: it reproduces on a random sample five times larger. Out of sample, the defensible statement is that federated and pooled training both improve the ranking of anomalous timesteps over the typical client, and that neither is shown to improve the archive’s own localization accuracy.

G. Counterfactual quality

The method also returns an explanation: the tokens it holds responsible are masked, resampled from the prior with the remaining tokens as context, and decoded back to a waveform, as Fig. 2(c) shows on one client, there under the deployed rule, whose mask is whatever the client’s own threshold declares. This is the only output that passes through the decoder, local in every configuration, so whether federation costs anything here is a separate question. The comparison below is instead paired on the window and on the rewritten set $\mathcal{M}$, which is label-derived so that $|\mathcal{M}|$ cannot vary with the arm: eight anomalous windows and eight normal ones of matching $|\mathcal{M}|$, on each of five series. Plausibility is the relative gain $\Delta = [\text{NLL}(x) - \text{NLL}(x_{cf})]/\text{NLL}(x)$ on the rewritten positions, with x_{cf} re-encoded from the waveform since scoring the sampled tokens would be circular, and specificity is $G = \Delta(\text{anomalous}) -$

TABLE IV

COUNTERFACTUAL QUALITY. Δ = PLAUSIBILITY GAIN ON THE REWRITTEN POSITIONS, EACH ARM UNDER ITS OWN PRIOR; G = SPECIFICITY GAP BETWEEN ANOMALOUS AND NORMAL WINDOWS. ABOVE THE RULE, THE FOUR DISCRIMINATING SERIES.

series	Δ plausibility			G specificity		
	local	centr.	(g)	local	centr.	(g)
ucr_011	0.541	0.577	0.588	0.289	0.609	0.403
ucr_014	0.411	0.542	0.600	0.232	0.263	0.270
ucr_043	0.264	0.166	0.232	0.083	0.110	0.213
ucr_170	0.415	0.559	0.225	0.406	0.685	0.553
ucr_001	0.286	0.208	0.295	0.073	−0.015	0.118

$\Delta(\text{normal})$, which controls for a generator that makes every input more typical. Each arm is judged by its own prior, so the levels of Δ are not comparable across arms while G is.

Every arm rewrites anomalies into content its own prior finds more typical than what it replaced ($\Delta > 0$ in all 15 cells of Table IV) and all but one pass the specificity control ($G > 0$ in 14 of 15, the exception being centralized training on ucr_001 at -0.015), so the repairs appear targeted rather than the result of uniform smoothing. On the comparable quantity federation does not lose: specificity favors it over local training on all five series, median paired difference $+0.114$, and is level with centralized training at $+0.007$. Five series, one seed and no counterfactual ground truth in UCR make this a null rather than evidence of equivalence, but within those limits federation shows no loss comparable to the differences it produces in detection.

H. Limitations

Every cell is a single seed, so per-series differences below a few hundredths should not be read as effects. Six of the ten development series cannot separate the arms at all, and that set was chosen purposively, so no quantity measured on it estimates a population; that role belongs to the random sample, whose primary endpoint proved too coarse to separate any pair of arms. Finally, the aligned and unaligned branches of Section VI-D were not run entirely on matched hardware, (g) having run on one accelerator while (e) used two others for seven of the ten series: a confound we can declare but not bound.

VII. CONCLUSION

We asked what a federated learner should share when the detector is built from a discrete tokenizer and a prior over the tokens it emits, and the two stages turn out not to be separable choices. Sharing the body of the prior improves detection only after the clients’ encoders have been averaged, and the identical modification is a null when they have not been. Alignment here is not a label but a quantity read off the checkpoints: under the strongest configuration the five clients tokenize identically, against a median $\kappa = 0.07$ with local encoders and a bit-identical codebook on both sides. Within that constraint, *what* is shared matters more than *how* it is aggregated: replacing encoder FedAvg by FedProx, by FedProto or by a control that shares only the initialization

changes nothing measurable, while changing the codebook rule changes a great deal on individual series and nothing in aggregate.

The empirical bottom line is deliberately narrow. Out of sample the federated model improves the ranking of anomalous timesteps over local training and matches pooled training on the archive’s own localization metric, on which our pre-registered primary endpoint separates no pair of arms at all, pooled training included: what the sample establishes is parity with an upper bound that does not itself separate. Weight-space aggregation can also propagate a false alarm raised by a minority of the clients to all of them, and there fusing the local models’ scores instead of their weights recovers the gap at no training cost, the diversity an ensemble exploits being exactly what parameter averaging destroys. On the counterfactual, federation costs nothing we can measure, which locates its price entirely in detection.

Repeating the matched contrasts across seeds would turn the interaction into a quantity, and the alignment axis, measured only at its extremes, leaves open where partial mechanisms land. Our split is also deliberately homogeneous, and two choices are expected to invert with heterogeneous clients: pooling the BN statistics, adopted against the FedBN prescription [25] precisely because no feature shift exists here, and a single shared codebook.

DATA AND RESOURCES

The data used in this study are in the UCR Time Series Anomaly Archive. The implementation is available at [REPOSITORY URL].

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